

SCm Vacuum Manifold — Primordial First-Principle Derivation

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April 24, 2026

1 SCm Vacuum Manifold (Pre-Gravity First Principle)

The primordial “matter” is the superconducting vacuum manifold:

$$\rho_{\text{vac,SCm}} = \rho_{\text{vacSCm}}$$

Negative-time modulation:

$$\cos(\pi t_n) = \cos(\pi t_n)$$

Buoyancy force (long-form integral):

$$\int_0^\infty \left(-F_0 + \frac{GM}{r^2} + U_{UA} \rho_{\text{vacSCm}} \cos(\pi t_n) \right) dr$$

26D Vacuum Density Series:

$$\sum_{n=1}^{\infty} \frac{\left(\frac{57}{100}\right)^n}{n^{26}}$$

Phonon activation at 1.25 THz:

$$e^{-\frac{(\omega - 125000000000.0)^2}{2\Gamma^2}}$$

Primordial split via $E_{\text{net}}(t, \Gamma)$:

$$E_{\text{net}}(t_n, \Gamma)$$

99-system master buoyancy sum ($k = 1 \dots 99$):

$$\left(-\frac{59.4MUg_k}{r^2} + 2.75 \cdot 10^{-7} \right) \cos(\pi t_n)$$

2 Monte Carlo Validation

$\langle F_{U,Bi,i} \rangle = 5.937e+05$ N, $\sigma = 3.784e+07$ N, 90% CI = $[-5.297e+07, 5.303e+07]$ N. Progress: 87%.

3 Comparison to Holmlid Rydberg Matter

Ultra-dense D(-1): $d = 2.3 \pm 0.1$ pm, density 10^{29} cm $^{-3}$, KER = 630 eV. SCm supplies the stabilizing $F_{U,Bi,i}$ and phonon background.

4 Refined Proof Fragment Outline (Riemann + Cosmogenesis)

1. SCm Vacuum Manifold (1 page)
2. Primordial Split & Cosmic Quantum Egg (0.5 page)
3. Holmlid Rydberg Bridge (0.5 page)
4. Riemann Closure via SCm (1 page)
5. Numerical Validation (0.5–1 page)